

Test 30: Dirac cone vF scaling vs L

Verdict: PASS | Generated: 2026-09-02 18:58:25 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j0+1]-ev[j0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

METHOD: EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j0+1]-ev[j0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

Test 30 HARDCORE: 11-point fit $R?=0.9999$, $v_F=1.9447$, $E?L$ cv=0.75%, zero tower 4 ?L.

Result [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 30 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[18:47:06.239] [+17941.521s] build_ab_cloud_hamiltonian: 16x16 (N=256), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:47:08.810] [+17944.092s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

FULL COMPUTATION LOG

[18:47:16.759] [+17946.041s] build_ab_cloud_hamiltonian: 32x32 (N=1024), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:47:13.274] [+17948.556s] build_ab_cloud_hamiltonian: 40x40 (N=1600), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:47:16.985] [+17952.268s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:47:23.131] [+17958.413s] build_ab_cloud_hamiltonian: 56x56 (N=3136), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:47:35.865] [+17971.147s] build_ab_cloud_hamiltonian: 64x64 (N=4096), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:48:01.478] [+17996.760s] build_ab_cloud_hamiltonian: 72x72 (N=5184), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:48:52.726] [+18048.009s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:50:38.784] [+18154.066s] build_ab_cloud_hamiltonian: 88x88 (N=7744), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[18:53:39.520] [+18334.803s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

??

TEST 30 (pass 2, HARDCORE): Dirac cone velocity ? dense-grid audit

??

CONSOLE CAPTURE

11-point fit $R?=0.9999$, $v_F=1.9447$, $E?L$ cv=0.75%, zero tower 4 ?L

? RUNTIME: 11 value-only diagonalizations, L up to 96 ? expect ? 8?12 min

L=16: E? = 0.765367 (E?•L = 12.2459), zero_modes = 4, 0.7 s

L=24: E? = 0.517638 (E?•L = 12.4233), zero_modes = 4, 0.7 s

L=32: E? = 0.390181 (E?•L = 12.4858), zero_modes = 4, 1.1 s

L=40: E? = 0.312869 (E?•L = 12.5148), zero_modes = 4, 2.5 s

L=48: E? = 0.261052 (E?•L = 12.5305), zero_modes = 4, 4.9 s

L=56: E? = 0.223929 (E?•L = 12.5400), zero_modes = 4, 11.4 s

L=64: E? = 0.196034 (E?•L = 12.5462), zero_modes = 4, 24.2 s

L=72: E? = 0.174311 (E?•L = 12.5504), zero_modes = 4, 49.8 s

L=80: E? = 0.156918 (E?•L = 12.5535), zero_modes = 4, 100.7 s

L=88: E? = 0.142678 (E?•L = 12.5557), zero_modes = 4, 176.0 s

L=96: E? = 0.130806 (E?•L = 12.5574), zero_modes = 4, 282.4 s

Fit: gap = 12.2188/L + 0.0054, R? = 0.999880, v_F = 1.9447 (?-flux target 2.0)

E?•L: mean = 12.5003, cv = 0.75% ? ? 5%

??

R? > 0.95 (11 pts) PASS ? R? = 0.9999

zero tower == 4 ?L PASS ? |E| ? 1e-8 at every size

E?•L cv ? 5% PASS ? cv = 0.75% (Dirac law: E?•L ? 4? = const)

Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS